

B. STAAD INPUT DATA

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*****
*          STAAD.Pro          *
*          Version 2007      Build 01          *
*          Proprietary Program of              *
*          Research Engineers, Intl.            *
*          Date=    OCT 14, 2008                *
*          Time=    16:25: 8                    *
*          USER ID: ENGINEERING                *
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1. STAAD SPACE
INPUT FILE: SKYLIGHT-GLASS_2476 X 1933.STD
2. START JOB INFORMATION
3. JOB NAME GLASS ANALYSIS - TYPICAL DIMENSION
4. JOB CLIENT DMP
5. JOB COMMENT FINAL
6. ENGINEER NAME AA
7. JOB REV 2
8. ENGINEER DATE 17-MAR-08
9. JOB NO 101
10. END JOB INFORMATION
11. INPUT WIDTH 79
12. UNIT MMS NEWTON
13. JOINT COORDINATES
14. 1 0 0 0; 2 0 0 1933; 3 2475 0 1933; 4 1965 0 0; 5 115 0 0; 6 115 0 1933
15. 7 0 0 115; 8 115 0 115; 9 0 0 1818; 10 115 0 1818; 11 2444.66 0 1818
16. 12 1995.34 0 115; 13 1715 0 0; 14 1715 0 1933; 15 1715 0 115; 16 1715 0 1818
17. 17 115 0 1704.47; 18 0 0 1704.47; 19 115 0 1590.93; 20 0 0 1590.93
18. 21 115 0 1477.4; 22 0 0 1477.4; 23 115 0 1363.87; 24 0 0 1363.87
19. 25 115 0 1250.33; 26 0 0 1250.33; 27 115 0 1136.8; 28 0 0 1136.8
20. 29 115 0 1023.27; 30 0 0 1023.27; 31 115 0 909.733; 32 0 0 909.733
21. 33 115 0 796.2; 34 0 0 796.2; 35 115 0 682.667; 36 0 0 682.667
22. 37 115 0 569.133; 38 0 0 569.133; 39 115 0 455.6; 40 0 0 455.6
23. 41 115 0 342.067; 42 0 0 342.067; 43 115 0 228.533; 44 0 0 228.533
24. 45 229.286 0 115; 46 229.286 0 0; 47 343.571 0 115; 48 343.571 0 0
25. 49 457.857 0 115; 50 457.857 0 0; 51 572.143 0 115; 52 572.143 0 0
26. 53 686.429 0 115; 54 686.429 0 0; 55 800.714 0 115; 56 800.714 0 0
27. 57 915 0 115; 58 915 0 0; 59 1029.29 0 115; 60 1029.29 0 0; 61 1143.57 0 115
28. 62 1143.57 0 0; 63 1257.86 0 115; 64 1257.86 0 0; 65 1372.14 0 115
29. 66 1372.14 0 0; 67 1486.43 0 115; 68 1486.43 0 0; 69 1600.71 0 115
30. 70 1600.71 0 0; 71 229.286 0 1933; 72 229.286 0 1818; 73 343.571 0 1933
31. 74 343.571 0 1818; 75 457.857 0 1933; 76 457.857 0 1818; 77 572.143 0 1933
32. 78 572.143 0 1818; 79 686.429 0 1933; 80 686.429 0 1818; 81 800.714 0 1933
33. 82 800.714 0 1818; 83 915 0 1933; 84 915 0 1818; 85 1029.29 0 1933
34. 86 1029.29 0 1818; 87 1143.57 0 1933; 88 1143.57 0 1818; 89 1257.86 0 1933
35. 90 1257.86 0 1818; 91 1372.14 0 1933; 92 1372.14 0 1818; 93 1486.43 0 1933
36. 94 1486.43 0 1818; 95 1600.71 0 1933; 96 1600.71 0 1818; 97 229.286 0 1704.47
37. 98 343.571 0 1704.47; 99 457.857 0 1704.47; 100 572.143 0 1704.47
38. 101 686.429 0 1704.47; 102 800.714 0 1704.47; 103 915 0 1704.47
39. 104 1029.29 0 1704.47; 105 1143.57 0 1704.47; 106 1257.86 0 1704.47
40. 107 1372.14 0 1704.47; 108 1486.43 0 1704.47; 109 1600.71 0 1704.47
41. 110 1715 0 1704.47; 111 229.286 0 1590.93; 112 343.571 0 1590.93
42. 113 457.857 0 1590.93; 114 572.143 0 1590.93; 115 686.429 0 1590.93
43. 116 800.714 0 1590.93; 117 915 0 1590.93; 118 1029.29 0 1590.93
44. 119 1143.57 0 1590.93; 120 1257.86 0 1590.93; 121 1372.14 0 1590.93
45. 122 1486.43 0 1590.93; 123 1600.71 0 1590.93; 124 1715 0 1590.93
46. 125 229.286 0 1477.4; 126 343.571 0 1477.4; 127 457.857 0 1477.4
47. 128 572.143 0 1477.4; 129 686.429 0 1477.4; 130 800.714 0 1477.4
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COMPLETE JOINT COORDINATES WILL BE PROVIDED IN SOFT COPY UPON REQUEST.

139. ELEMENT INCIDENCES SHELL

140. 1 7 8 5 1; 2 2 6 10 9; 3 9 10 17 18; 4 18 17 19 20; 5 20 19 21 22
 141. 6 22 21 23 24; 7 24 23 25 26; 8 26 25 27 28; 9 28 27 29 30; 10 30 29 31 32
 142. 11 32 31 33 34; 12 34 33 35 36; 13 36 35 37 38; 14 38 37 39 40; 15 40 39 41 42
 143. 16 42 41 43 44; 17 44 43 8 7; 18 8 45 46 5; 19 45 47 48 46; 20 47 49 50 48
 144. 21 49 51 52 50; 22 51 53 54 52; 23 53 55 56 54; 24 55 57 58 56; 25 57 59 60 58
 145. 26 59 61 62 60; 27 61 63 64 62; 28 63 65 66 64; 29 65 67 68 66; 30 67 69 70 68
 146. 31 69 15 13 70; 32 6 71 72 10; 33 71 73 74 72; 34 73 75 76 74; 35 75 77 78 76
 147. 36 77 79 80 78; 37 79 81 82 80; 38 81 83 84 82; 39 83 85 86 84; 40 85 87 88 86
 148. 41 87 89 90 88; 42 89 91 92 90; 43 91 93 94 92; 44 93 95 96 94; 45 95 14 16 96
 149. 46 10 72 97 17; 47 72 74 98 97; 48 74 76 99 98; 49 76 78 100 99
 150. 50 78 80 101 100; 51 80 82 102 101; 52 82 84 103 102; 53 84 86 104 103
 151. 54 86 88 105 104; 55 88 90 106 105; 56 90 92 107 106; 57 92 94 108 107
 152. 58 94 96 109 108; 59 96 16 110 109; 60 17 97 111 19; 61 97 98 112 111
 153. 62 98 99 113 112; 63 99 100 114 113; 64 100 101 115 114; 65 101 102 116 115
 154. 66 102 103 117 116; 67 103 104 118 117; 68 104 105 119 118; 69 105 106 120 119
 155. 70 106 107 121 120; 71 107 108 122 121; 72 108 109 123 122; 73 109 110 124 123
 156. 74 19 111 125 21; 75 111 112 126 125; 76 112 113 127 126; 77 113 114 128 127
 157. 78 114 115 129 128; 79 115 116 130 129; 80 116 117 131 130; 81 117 118 132 131
 158. 82 118 119 133 132; 83 119 120 134 133; 84 120 121 135 134; 85 121 122 136 135
 159. 86 122 123 137 136; 87 123 124 138 137; 88 21 125 139 23; 89 125 126 140 139
 160. 90 126 127 141 140; 91 127 128 142 141; 92 128 129 143 142; 93 129 130 144 143
 161. 94 130 131 145 144; 95 131 132 146 145; 96 132 133 147 146; 97 133 134 148 147
 162. 98 134 135 149 148; 99 135 136 150 149; 100 136 137 151 150
 163. 101 137 138 152 151; 102 23 139 153 25; 103 139 140 154 153
 164. 104 140 141 155 154; 105 141 142 156 155; 106 142 143 157 156
 165. 107 143 144 158 157; 108 144 145 159 158; 109 145 146 160 159
 166. 110 146 147 161 160; 111 147 148 162 161; 112 148 149 163 162
 167. 113 149 150 164 163; 114 150 151 165 164; 115 151 152 166 165
 168. 116 25 153 167 27; 117 153 154 168 167; 118 154 155 169 168
 169. 119 155 156 170 169; 120 156 157 171 170; 121 157 158 172 171
 170. 122 158 159 173 172; 123 159 160 174 173; 124 160 161 175 174
 171. 125 161 162 176 175; 126 162 163 177 176; 127 163 164 178 177
 172. 128 164 165 179 178; 129 165 166 180 179; 130 27 167 181 29
 173. 131 167 168 182 181; 132 168 169 183 182; 133 169 170 184 183
 174. 134 170 171 185 184; 135 171 172 186 185; 136 172 173 187 186
 175. 137 173 174 188 187; 138 174 175 189 188; 139 175 176 190 189
 176. 140 176 177 191 190; 141 177 178 192 191; 142 178 179 193 192
 177. 143 179 180 194 193; 144 29 181 195 31; 145 181 182 196 195
 178. 146 182 183 197 196; 147 183 184 198 197; 148 184 185 199 198
 179. 149 185 186 200 199; 150 186 187 201 200; 151 187 188 202 201
 180. 152 188 189 203 202; 153 189 190 204 203; 154 190 191 205 204
 181. 155 191 192 206 205; 156 192 193 207 206; 157 193 194 208 207
 182. 158 31 195 209 33; 159 195 196 210 209; 160 196 197 211 210
 183. 161 197 198 212 211; 162 198 199 213 212; 163 199 200 214 213
 184. 164 200 201 215 214; 165 201 202 216 215; 166 202 203 217 216
 185. 167 203 204 218 217; 168 204 205 219 218; 169 205 206 220 219
 186. 170 206 207 221 220; 171 207 208 222 221; 172 33 209 223 35
 187. 173 209 210 224 223; 174 210 211 225 224; 175 211 212 226 225
 188. 176 212 213 227 226; 177 213 214 228 227; 178 214 215 229 228
 189. 179 215 216 230 229; 180 216 217 231 230; 181 217 218 232 231
 190. 182 218 219 233 232; 183 219 220 234 233; 184 220 221 235 234
 191. 185 221 222 236 235; 186 35 223 237 37; 187 223 224 238 237
 192. 188 224 225 239 238; 189 225 226 240 239; 190 226 227 241 240
 193. 191 227 228 242 241; 192 228 229 243 242; 193 229 230 244 243
 194. 194 230 231 245 244; 195 231 232 246 245; 196 232 233 247 246
 195. 197 233 234 248 247; 198 234 235 249 248; 199 235 236 250 249
 196. 200 37 237 251 39; 201 237 238 252 251; 202 238 239 253 252
 197. 203 239 240 254 253; 204 240 241 255 254; 205 241 242 256 255
 198. 206 242 243 257 256; 207 243 244 258 257; 208 244 245 259 258

COMPLETE ELEMENT INCIDENCES SHELL WILL BE PROVIDED IN SOFT COPY UPON REQUEST.

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270. ELEMENT PROPERTY
271. 1 TO 482 THICKNESS 24
272. DEFINE MATERIAL START
273. ISOTROPIC GLASS
274. E 72000
275. POISSON 0.22
276. DENSITY 2.4525E-005
277. ALPHA 8.5E-006
278. DAMP 0.03
279. END DEFINE MATERIAL
280. CONSTANTS
281. MATERIAL GLASS ALL
282. SUPPORTS
283. 8 PINNED
284. 3 4 11 12 293 TO 306 FIXED BUT MX MZ
285. 10 FIXED BUT FZ MX MY MZ
286. 16 FIXED BUT FX FZ MX MY MZ
287. LOAD 1 LOADTYPE MASS TITLE SELFWEIGHT
288. SELFWEIGHT Y -1
289. LOAD 2 LOADTYPE LIVE TITLE CASE ONE_POINT LOAD
290. JOINT LOAD
291. 113 118 124 197 202 208 267 272 292 FY -500
292. LOAD 3 LOADTYPE LIVE TITLE CASE TWO_PLATE LOAD
293. ELEMENT LOAD
294. 1 TO 482 PR GY -0.001
295. LOAD 4 LOADTYPE TEMPERATURE TITLE THERMAL LOAD
296. TEMPERATURE LOAD
297. 1 TO 482 TEMP 0 32 32
298. LOAD 5 LOADTYPE WIND TITLE WIND LOAD_POS
299. ELEMENT LOAD
300. 1 TO 482 PR GY 0.00152
301. LOAD 6 LOADTYPE WIND TITLE WIND LOAD_NEG
302. ELEMENT LOAD
303. 1 TO 482 PR GY -0.00292
304. LOAD COMB 7 (SLF + POINT LOAD)_NOMINAL
305. 1 1.0 2 1.0
306. LOAD COMB 8 (SLF + PLATE LOAD)_NOMINAL
307. 1 1.0 3 1.0
308. LOAD COMB 9 (SLF + POS_WL)_NOMINAL
309. 1 1.0 5 1.0
310. LOAD COMB 10 (SLF + NEG_WL)_NOMINAL
311. 1 1.0 6 1.0
312. LOAD COMB 11 (SLF + POINT LOAD + THERMAL LOAD)_FACTORED
313. 1 1.4 2 1.4 4 1.0
314. LOAD COMB 12 (SLF + PLATE LOAD + THERMAL LOAD)_FACTORED
315. 1 1.4 3 1.4 4 1.0
316. LOAD COMB 13 (SLF + WIND LOAD_POS + THERMAL LOAD)_FACTORED
317. 1 1.4 5 1.4 4 1.0
318. LOAD COMB 14 (SLF + WIND LOAD_NEG + THERMAL LOAD)_FACTORED
319. 1 1.4 6 1.4 4 1.0
320. PERFORM ANALYSIS
321. PRINT MAX. JOINT DISPLACEMENT
JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

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JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
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60	10	0.0000	-0.6144	0.0000	0.0010	0.0000	-0.0007
229	10	0.0000	-0.6954	0.0000	0.0003	0.0000	-0.0004